

TEN YEARS OF MICROTRONIC EMULATORS

Some people build ships in bottles. Over 2016–2026, Michael Wessel rebuilt the 1981 **Busch Microtronic** in hardware — a family of Arduino-based emulators, from the first **Mega 2560** to the **Phoenix** that runs the original 1981 firmware ROM — winning two international retro-computing prizes along the way.

★ **Hackaday “Reinvented Retro” Contest 2021 — winner** (the Next-Generation Nokia-LCD build), and featured on the Hackaday **front page**.

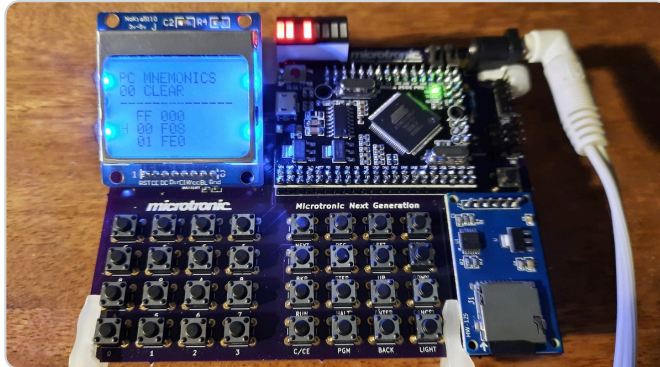
★ **RetroChallenge 2021/10 — prize winner**: a MIDI drum computer written in Microtronic machine code on the Retro-Authentic **Bubble-LED** Microtronic.

01 THE JOURNEY — A DECADE OF BUILDS



Mega 2560
2016 · THE ORIGINAL

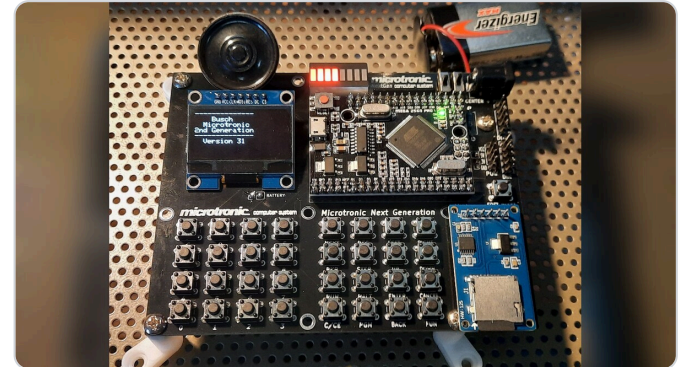
A whole Microtronic in an Arduino Mega + LCD, with Emic-2 speech synthesis. Where the road began.



Next Generation · Nokia LCD
2019–21 · PRIZE WINNER

Custom PCB: live-disassembler LCD, SD-card storage, 2095 cassette emulation, RTC, sound, DIN/DOT I/O.

★ **Hackaday front page + Reinvented Retro 2021 winner**



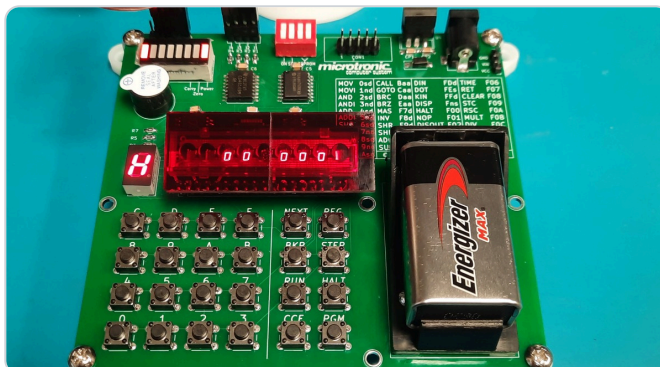
Next Generation · OLED
2021+ · THE REFINEMENT

The same design with a crisp 1.3" SH1106 OLED — big-font, register and disassembler display modes.



2nd Generation
2020 · SISTER BUILD

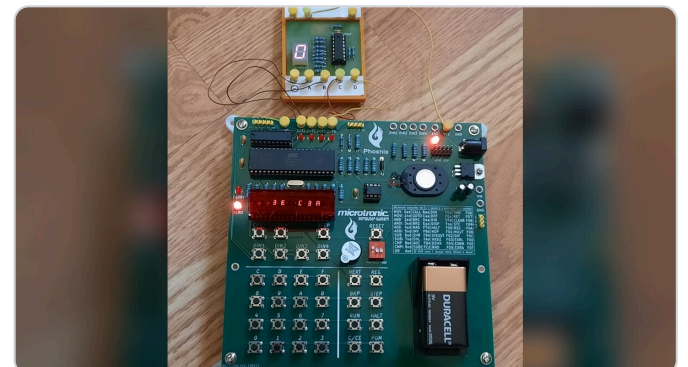
By Frank de Jaeger & Manfred Henf: same firmware, a 3D-printed keypad into a real Busch console, transistor-accurate I/O.



Retro-Authentic Bubble-LED
2021

A TI/TMS bubble-LED display for true 1981 authenticity; host to a MIDI drum computer written in Microtronic machine code.

★ **RetroChallenge 2021/10 winner**



Phoenix
2025 · THE FINALE

Runs the **actual 1981 firmware ROM** on new hardware — with Jason T. Jacques & Decle.

★ **Hackster & Hackaday featured**

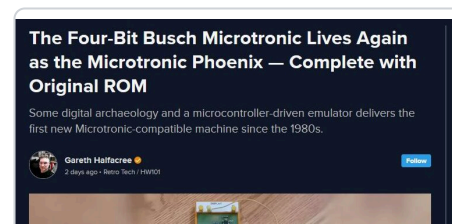
02 THE ARC — FROM EMULATING BEHAVIOUR TO RUNNING THE REAL ROM

For most of the journey the emulators reproduced the Microtronic's *behaviour*, worked out from its manuals — the 1981 firmware was mask-programmed *inside* the TI TMS1600 chip and had never been dumped. The finale changed that: together with **Jason T. Jacques** and **Decle**, the original ROM was finally extracted and disassembled, and the **Phoenix** now executes the **genuine 1981 firmware** on a cycle-accurate TMS1600 emulator (alongside Michael's own Arduino “Neo” re-implementation). The real machine's soul, reborn on new silicon.

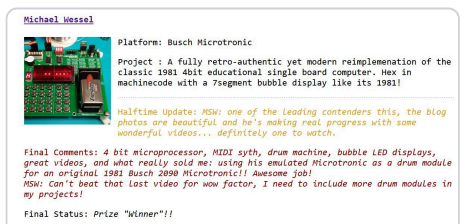
2016 Mega 2560 → 2019 Next-Gen LCD → 2020 2nd Gen → 2021 Bubble-LED + OLED → 2025 PHOENIX — the original ROM lives again

03 PRESS & PRIZES

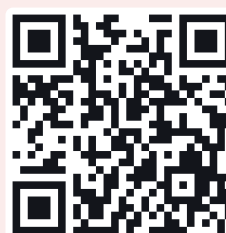
- ▶ **Hackaday “Reinvented Retro” Contest 2021 — winner** (Next Generation), and front-paged on Hackaday.
- ▶ **RetroChallenge 2021/10 — prize winner**: the Bubble-LED Microtronic MIDI drum computer.
- ▶ **Hackster.io** — “The Four-Bit Busch Microtronic Lives Again as the Microtronic Phoenix — Complete with Original ROM.”
- ▶ **Hackaday** — the Phoenix and the “firmware-ROM archaeology” project, featured.
- ▶ **LOAD #11** (Oct 2025) — Germany's VzEKC “classic computing” magazine covered the Microtronic and its emulators.
- ▶ Grew from **1 to 6 contributors** over ten years; full build videos on YouTube.



Hackster.io — the Phoenix



RetroChallenge 2021/10 — “Prize Winner”



All builds, firmware & Gerbers

Sketches, PCBs and docs for every emulator generation.

github.com/lambdamikel/Busch-2090

A decade-long project by **Michael Wessel** (2016–2026). The **2nd Generation** sister build is by **Frank de Jaeger & Manfred Henf**; the **Phoenix** firmware recovery and TMS1600 emulator are by **Jason T. Jacques & Decle** (with Michael's Arduino re-implementation, PCB & integration). 2095 cassette protocol reverse-engineered by Martin Sauter. Repos: github.com/lambdamikel/Busch-2090 · [microtronic-phoenix](https://github.com/microtronic-phoenix). Poster by Claude.